

Part 1:

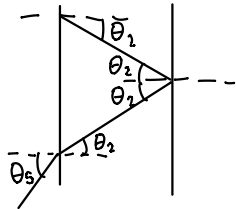
a) $y = z + \frac{z}{\cos \theta_2} \rightarrow z = \frac{y \cos \theta_2}{2 +}$

b) $x = \sin \theta_1, \frac{\partial x}{\partial \theta_1} = \cos \theta_1 \rightarrow \alpha_x = \cos \theta_1 \alpha_{\theta_1}$

$\frac{\partial z}{\partial y} = \frac{\cos \theta_2}{2 +}, \frac{\partial z}{\partial t} = -\frac{y \cos \theta_2}{2 +^2}, \alpha_z = \sqrt{\left(\frac{\cos \theta_2}{2 +} \alpha_y\right)^2 + \left(\frac{y \cos \theta_2}{2 +^2} \alpha_t\right)^2}$

c) $\theta_1 = \theta_4$ follows immediately from the law of reflection.

By Snell's Law, $\sin \theta_1 = n_{\text{block}} \sin \theta_2$ and $n_{\text{block}} \sin(\theta_2) = \sin \theta_3$. Since $\sin \theta_1 = \sin \theta_3$ for $\theta_1, \theta_3 \in [0, \frac{\pi}{2}]$, we must have $\theta_1 = \theta_3$. By the Law of Reflection, the angle of incidence as the light leaves the block is θ_2 . Then, by Snell's Law once again, $n_{\text{block}} \sin \theta_2 = \sin \theta_3$. By the same logic as before, this means $\theta_1 = \theta_3 = \theta_4$.



d) By Snell's Law, $x = n_{\text{block}} z \rightarrow n_{\text{block}} = \frac{x}{z}$

$z = \sin \theta_2 = \frac{\frac{1}{2} y}{\sqrt{1 + \frac{1}{4} y^2}}, \text{ so } \sin \theta_1 = n_{\text{block}} \cdot \frac{y}{2 \sqrt{1 + \frac{1}{4} y^2}} \rightarrow n_{\text{block}} = \frac{2 \sqrt{1 + \frac{1}{4} y^2} \sin \theta_1}{y}$

e) $\frac{\partial n_{\text{block}}}{\partial x} = \frac{1}{z}, \frac{\partial n_{\text{block}}}{\partial z} = -\frac{x}{z^2}, \alpha_n = \sqrt{\left(\frac{\alpha_x}{z}\right)^2 + \left(\frac{\alpha_z x}{z^2}\right)^2}$

Part 2:

a) From equation 5.3.4b of the statistics review sheet: $b = \frac{\sum_i w_i (y_i - x_i)}{\sum_i w_i}$

b) $\frac{\partial b}{\partial x_i} = \frac{-w_i}{\sum_j w_j}, \frac{\partial b}{\partial y_i} = \frac{w_i}{\sum_j w_j}, \alpha_b = \sqrt{\sum_i \left(\frac{w_i}{\sum_j w_j}\right)^2 (\alpha_{x_i}^2 + \alpha_{y_i}^2)}$

